

Thong (Tom) Huynh

 ThongHHuynh |  Thong Huynh |  Portfolio |  thong.huynh@ontariotechu.net

EDUCATION & AWARDS

Ontario Tech University — B.Eng., Mechatronics Engineering (GPA: 4.25/4.3) 2024–2028
Competitions — International Youth Robot Competition, MakeX, FIRST Robotics
Awards — Undergraduate Research Award, STAR Award, Merit Award, President's List

SKILLS

CAD and Design: SolidWorks Certified Professional, Siemens NX, Fusion 360, GD&T, DFM/DFA.
Robotics Hardware: NVIDIA Jetson Orin Nano/AGX, Raspberry Pi, depth cameras, LiDAR, IMUs, sensors, actuators.
Robotics and Simulation: ROS 2, Gazebo, NVIDIA Isaac Sim/Lab, Nav2.
Programming and Vision: Linux, Python, C++, OpenCV, PyTorch, YOLO, RF-DETR, DINO/DINOv2.

WORK EXPERIENCES

Robotics Software Engineer | ABI Ltd. May 2026 – Present
– Developed end-to-end digital twins of industrial robotic workcells in NVIDIA Isaac Sim to accelerate system validation.
– Architected an in-house, two-stage YOLO with CNN head for bakery inspection pipeline, achieving < 100 ms inference latency, cutting \$100K in outsourcing costs.
– Engineered an internal computer vision tooling suite adopted company-wide across the engineering team, improving data labeling and model evaluation workflows.
– Supporting the integration of 6-DOF ABB robotic arms for high-speed automated bread-scoring systems.

Research Assistant | MARS Lab May 2025 – May 2026
– Developed a hybrid multimodal walking-and-flying robotic platform, designed 10+ custom mechanical components to optimize structural rigidity and payload capacity.
– Formulated forward/inverse kinematics for quadruped locomotion and built an end-to-end Gazebo simulation pipeline for dynamic gait and motion validation.
– Implemented OctoMap-based 3D mapping, navigation, and obstacle avoidance via MAVROS.

Design Team Lead | Ontario Tech RoboMaster September 2024 – Present
– Directed the mechanical design of a 500+ component competition robot in SolidWorks, ensuring full DFM/DFA compliance for manufacturing.
– Managed CAD version control and conducted design reviews to maintain consistency across mechanical and robotic subsystems.
– Mentored team members in mechanical design, GD&T, engineering drawings, and robotics design principles.

INDEPENDENT PROJECTS

SensQ – Autonomous Mobile Robot Platform January 2026 – Present
– Built and deployed an autonomous mobile robot integrating physical hardware with ROS 2.
– Implemented Nav2 and SLAM Toolbox navigation, custom A* and Boustrophedon coverage planning, and AprilTag-based docking.

Bolt Eye – Automated Inspection System August 2025 – Present
– Designed an automated conveyor system integrated with custom optical inspection hardware.
– Deployed a real-time inspection pipeline (YOLOv11, Mask R-CNN, DINOv2) with sub-100 ms latency.
– Detected and classified surface defects down to ≤ 2 mm on active production lines.

6-DOF Robotic Arm May 2024 – January 2026
– Built a functional 6-DOF robotic arm with custom kinematics and L-BFGS-B path planning.
– Implemented Python and ROS/RViz tools for motion visualization and validation.